

Week 6 – NLP Self-Study

What I did

- I used 250 course messages.
- The goal was to predict the message category from the message text.
- There were 5 categories, with 50 messages in each.
- There were no missing messages.
- I found 40 duplicate messages.
- I used 187 messages for training and 63 for testing.

Results

- DummyClassifier: **19%**
- TF-IDF + Logistic Regression: **76%**
- TF-IDF + Naive Bayes: **76%**
- CountVectorizer + Logistic Regression: **84%**
- CountVectorizer with unigram + bigram: **86%**

What I found

- The real models performed much better than the 19% baseline.
- CountVectorizer worked better than TF-IDF in this dataset.
- Adding bigrams improved the result from 84% to 86%.
- Some messages were difficult because they contained words related to different categories.
- Duplicate messages are a limitation because similar messages could appear in both training and test data.

Conclusion

My best result was **86% accuracy** using CountVectorizer with unigrams and bigrams. The model worked quite well, but some messages were still difficult to classify correctly.